

Course Name: Big Data Analytics - II

Course Code: 05201353

Prerequisite: Basic Knowledge of Big Data Technologies and Frameworks

Rationale: The objective of this course is to provide Conceptual insight about Big Data using Map-reduce programming in Both Hadoop and Spark framework.

Course Learning Objective:

CLOBJ 1	Understand the fundamentals and core concepts of Big Data, including characteristics, architecture, and various big data analytics techniques.
CLOBJ 2	Explore and implement data storage and processing using the Hadoop ecosystem, including HDFS, MapReduce, Hive, Pig, and HBase.
CLOBJ 3	Explain, analyze, and apply the MapReduce programming model for processing large datasets in both Hadoop and Apache Spark frameworks.
CLOBJ 4	Develop Big Data applications using Apache Spark with various programming languages such as Python (PySpark), Scala, and Java.
CLOBJ 5	Analyze and implement graph processing algorithms (e.g., PageRank, Connected Components) using Spark GraphX.
CLOBJ 6	Design and deploy real-time data processing pipelines using Spark Streaming to handle live streaming data effectively.

Course Learning Outcomes:

CLO 1	Understand the fundamentals of various big data analytics techniques.
CLO 2	Understand and work on Hadoop Framework and eco systems. Explain and Analyse the Big Data using Map-reduce programming in Both Hadoop and Spark framework.
CLO 3	Demonstrate spark programming with different programming languages
CLO 4	Demonstrate the graph algorithms and live streaming data in Spark

Teaching and Examination Scheme										Total
Teaching Scheme				Examination Scheme						
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	1	2	-	5	20	20	20	60	30	150

SEE - Semester End Examination, T - Theory, P - Practical

Course Content		W - Weightage (%) , T - Teaching hours		
Sr.	Topics	W	T	
1	Understanding Analytics and Big Data: Comparing Reporting and Analysis, Types of Analytics, Points to Consider during Analysis, Developing an Analytic Team, Understanding Text Analytics	8	3	

2	Hadoop Eco System Pig: Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive: Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Define Functions. Hbase: HBasics, Concepts, Clients, Example, Hbase Versus RDBMS. Introduction to MongoDB – JSON - Terms used in RDBMS and MongoDB - Data Types in MongoDB – MongoDB Query Language.	15	7
3	Customizing MapReduce Execution and Implementing MapReduce Program Controlling MapReduce Execution with InputFormat, Reading Data with Custom RecordReader, Organizing Output Data with OutputFormats, Customizing Data with RecordWriter, Optimizing MapReduce Execution with Combiner, Controlling Reducer Execution with Partitioners, Customizing the MapReduce Execution in Terms of YARN, Implementing a MapReduce Program for Sorting Text Data.	11	5
4	Testing and Debugging MapReduce Applications Debugging Hadoop MapReduce Locally, Performing Unit Testing for MapReduce Applications, Performing Local Application Testing with Eclipse, Logging for Hadoop Testing, Application Log Processing, Defensive Programming in MapReduce.	13	6
5	MapReduce Based Machine Learning Module content: K-Means, PLANET, Parallel SVM, Association Rule Mining in MapReduce, Inverted Index, Page Ranking, Expectation Maximization, Bayesian Networks	15	8
6	Spark Framework Introduction to GPU Computing, CUDA Programming Model, CUDA API, Simple Matrix, Multiplication in CUDA, CUDA Memory Model, Shared Memory Matrix Multiplication, Additional CUDA API Features.	14	7
7	Data Analysis with Spark Shell Writing Spark Application - Spark Programming in Scala, Python, R, Java - Application Execution. Spark SQL and GraphX SQL Context – Importing and Saving data – Data frames – using SQL – GraphX overview – Creating Graph – Graph Algorithms.	14	7
8	Spark Streaming Overview – Errors and Recovery – Streaming Source – Streaming live data with spark	10	5